Numerical Question

An LTE OFDM system has a nominal bandwidth of 10 MHz, 1024 FFT/IFFT points, 600 data-carrying subcarriers, and a guard fraction of 0.07. If 16-QAM modulation is used, calculate the raw data rate, neglecting channel coding.

$$R = \frac{B}{L}\frac{L_d \log_2(M)}{1+G}$$
$$= \frac{10^7 \text{MHz}}{1024}\frac{600 \log_2(16)}{1.07} = 21.9 \,\text{Mbps}.$$

Numerical Question — 20 MHz LTE

An LTE OFDM system has a nominal bandwidth of 20 MHz, 2048 FFT/IFFT points, 1200 data-carrying subcarriers, and a guard fraction of 0.07. If 16-QAM modulation is used, calculate the raw data rate, neglecting channel coding.

$$R = \frac{B}{L}\frac{L_d \log_2(M)}{1+G}$$

Since

$$\log_2(16) = 4$$

therefore,

$$R = \frac{20 \times 10^6}{2048} \frac{1200 \times 4}{1.07}$$

$$\boxed{R \approx 43.8 \text{ Mbps}}$$

So the **raw data rate for the 20 MHz LTE system is approximately 43.8 Mbps**.

Question

For a 10 MHz LTE system, the nominal bandwidth is $B = 7.68$ MHz. The relation is

$$B = \frac{f_s}{2}$$

Calculate the sampling frequency f_s.

Answer

$$f_s = 2B$$

$$= 2(7.68)$$

$$\boxed{f_s = 15.36 \text{ MHz}}$$

Question:
For a 20 MHz LTE system, the sampling frequency is 30.72 MHz. Calculate the nominal bandwidth B, using

$$B = \frac{f_s}{2}$$

Answer:

$$B = \frac{30.72}{2} = \boxed{15.36 \text{ MHz}}$$

Question:

For a 20 MHz LTE system, the subcarrier spacing is $15\,\text{kHz}$. Using the relation

$$T_s = \frac{1}{\Delta f \times 2048}$$

calculate the sampling time.

Answer:

$$T_s = \frac{1}{15 \times 10^3 \times 2048}$$

$$\boxed{T_s = 32.55\,\text{ns}}$$

Question

For a 10 MHz LTE system:

$$L = 1024, \qquad G = 0.07$$

Calculate the number of guard/CP samples N_g.

Solution

Using:

$$N_g = GL$$

$$N_g = 0.07 \times 1024$$

$$\boxed{N_g \approx 72\ \text{samples}}$$

Question

For the LTE system in Table 3.1, $T_s = 32.55$ ns. Calculate the guard time for:

1. Normal CP: $144T_s$
2. Extended CP: $160T_s$

Solution

Normal CP

$$T_g = 144T_s$$

$$= 144(32.55\,ns)$$

$$\boxed{T_g \approx 4.69\,\mu s \approx 4.7\,\mu s}$$

Extended CP

$$T_g = 160T_s$$

$$= 160(32.55\,ns)$$

$$\boxed{T_g \approx 5.21\,\mu s \approx 5.2\,\mu s}$$

Question

For the 10 MHz LTE system:

$$L = 1024, \qquad N_g = 72, \qquad B = 7.68\,MHz$$

Calculate the OFDM symbol time T using:

$$T = \frac{L + N_g}{B}$$

Solution

$$T = \frac{1024 + 72}{7.68 \times 10^6}$$

$$T = \frac{1096}{7.68 \times 10^6}$$

$$\boxed{T = 142.7\,\mu s}$$

Question:
For a 20 MHz LTE system, the FFT size is 2048 and the guard fraction is $G = 0.07$. Find the number of guard samples.

Solution:

$$N_g = GL$$

$$N_g = 0.07(2048)$$

$$\boxed{N_g \approx 143 \text{ samples}}$$

For the standard LTE normal CP value, this is conventionally represented as **144 samples**.

Question

An LTE OFDM system has:

$$f_s = 15.36\,MHz$$

and

$$\Delta f = 15\,kHz$$

Using the relation

$$L = \frac{f_s}{\Delta f}$$

calculate the FFT/IFFT size.

Solution

$$L = \frac{15.36 \times 10^6}{15 \times 10^3}$$

$$\boxed{L = 1024}$$

Question

An LTE OFDM system uses an FFT size of $L = 1024$ and a sampling frequency of 15.36 MHz. Find the subcarrier spacing.

Solution

$$\Delta f = \frac{f_s}{L}$$

$$= \frac{15.36 \times 10^6}{1024}$$

$$\boxed{\Delta f = 15\,kHz}$$

Question

A 10 MHz LTE system has $L = 1024$ FFT bins and $L_d = 600$ data-carrying subcarriers. How many FFT bins are not used for data?

Solution

$$N_{\text{remaining}} = L - L_d$$

$$= 1024 - 600$$

$$\boxed{424}$$

Question

For a 10 MHz LTE OFDM system, the following parameters are given:

$$B = 7.68\,MHz$$

$$L = 1024$$

$$G = 0.07$$

$$\Delta f = 15\,kHz$$

Calculate:

1. Number of guard samples N_g
2. Sampling time T_s
3. Normal CP duration $144T_s$
4. Extended CP duration $160T_s$
5. OFDM symbol time T

Solution

1. Guard samples

$$N_g = GL$$

$$= 0.07(1024)$$

$$\boxed{N_g = 72}$$

2. Sampling time

$$T_s = \frac{1}{15 \times 10^3 \times 2048}$$

$$\boxed{T_s = 32.55\,ns}$$

3. Normal CP

$$T_g = 144(32.55\,ns)$$

$$\boxed{T_g \approx 4.7\,\mu s}$$

4. Extended CP

$$T_g = 160(32.55\,ns)$$

$$\boxed{T_g \approx 5.2\,\mu s}$$

5. OFDM symbol time

$$T = \frac{L + N_g}{B}$$

$$= \frac{1024 + 72}{7.68 \times 10^6}$$

$$\boxed{T = 142.7\,\mu s}$$

Question

For a 10 MHz LTE system, the guard fraction is $G = 0.07$, and the number of guard samples is $N_g = 72$. Find the FFT/IFFT size L.

Solution

From:

$$N_g = GL$$

$$L = \frac{N_g}{G}$$

$$L = \frac{72}{0.07}$$

$$\boxed{L \approx 1024}$$